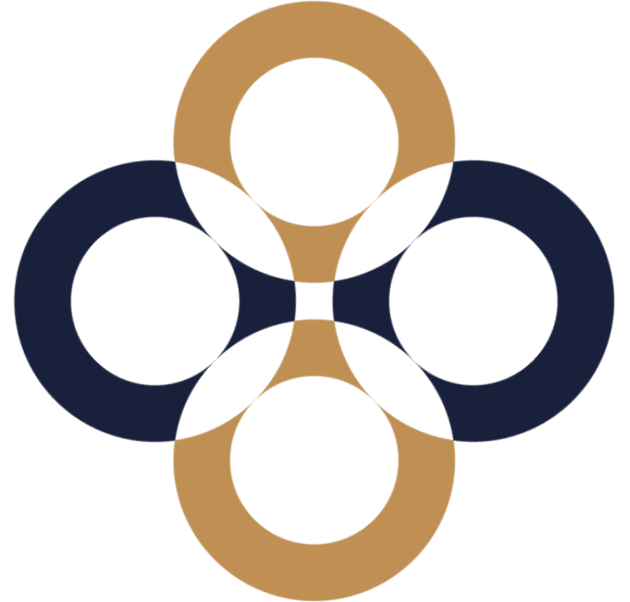
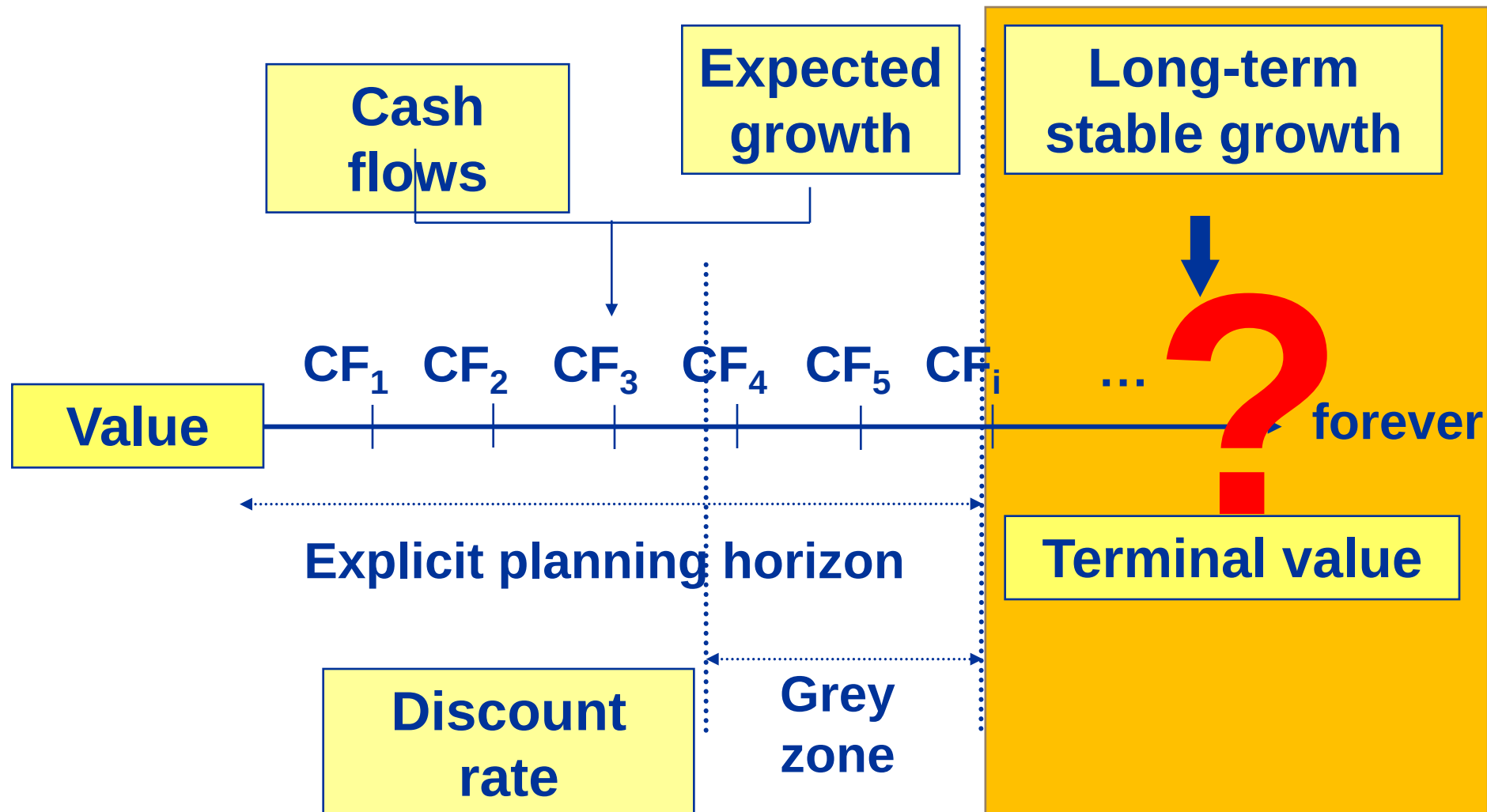


Estimating the terminal value

Péter Juhász, PhD, CFA

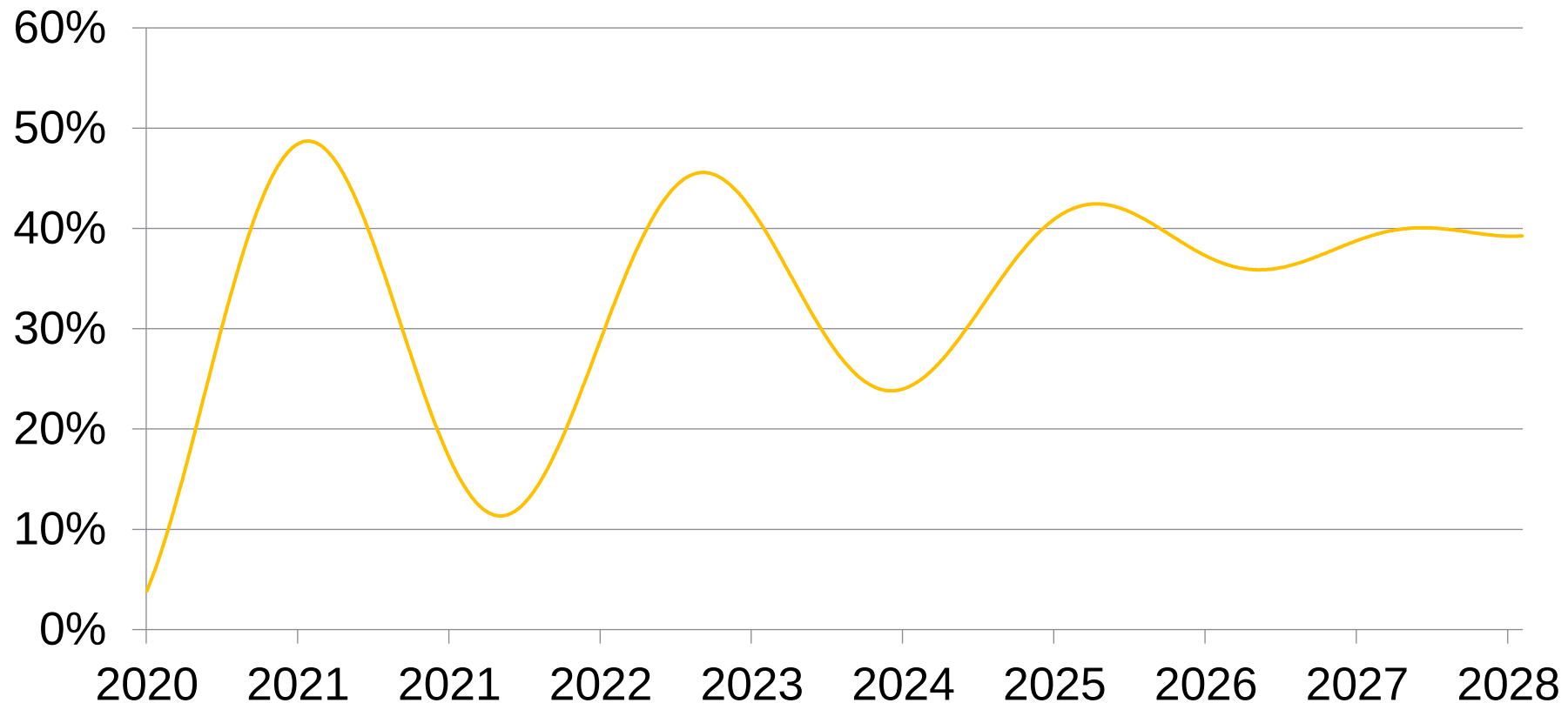


The general DCF model



How long should the explicit period last?

By the start of the terminal value period all inputs should stabilise.



The lifecycle of the ROIC and the g

Depend on the capabilities of the firm and the dynamics of the industry.

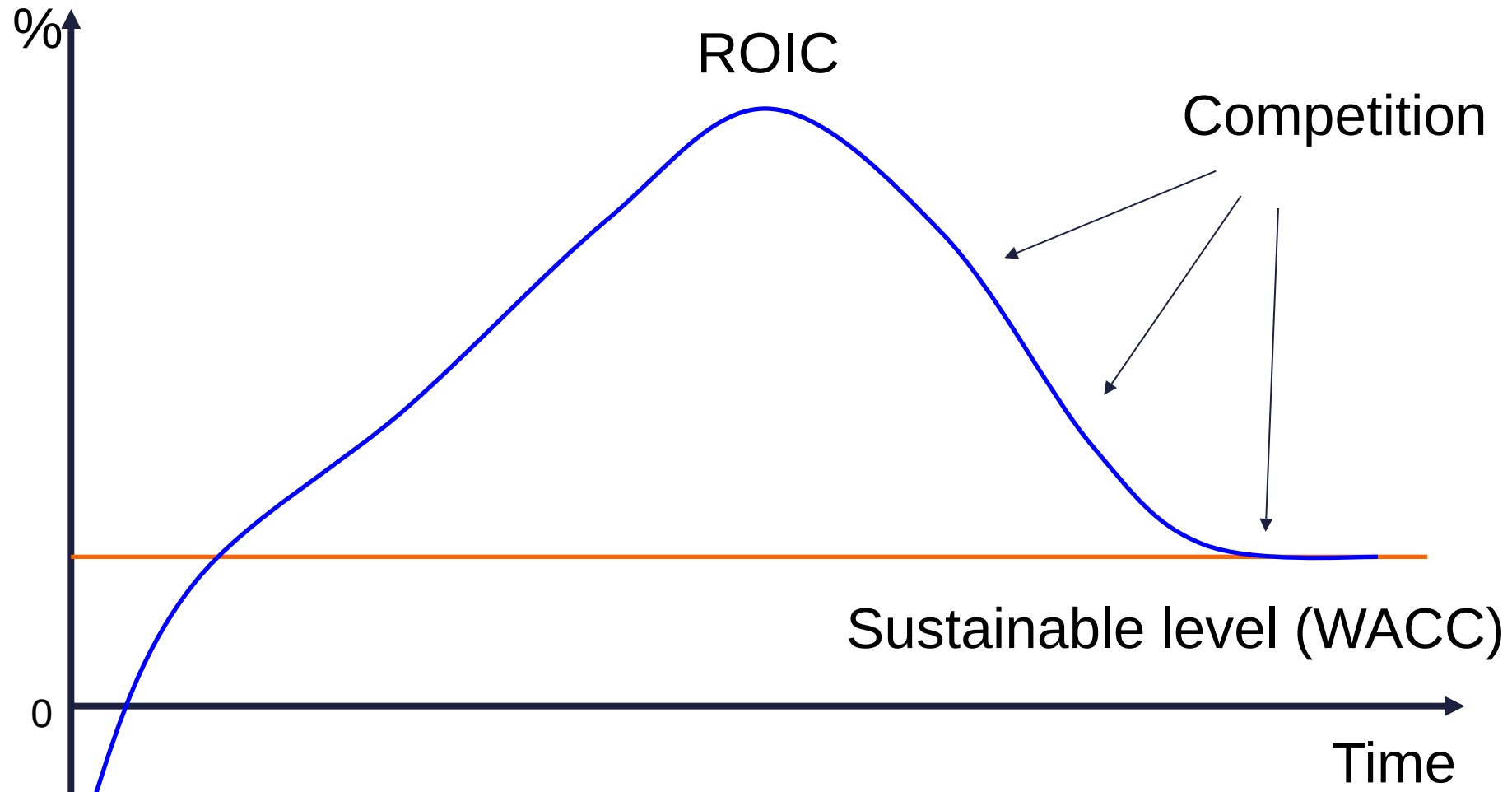
Those converge to the industry average.

Those mutually depend on each other (g generates value only if $ROIC > WACC$).

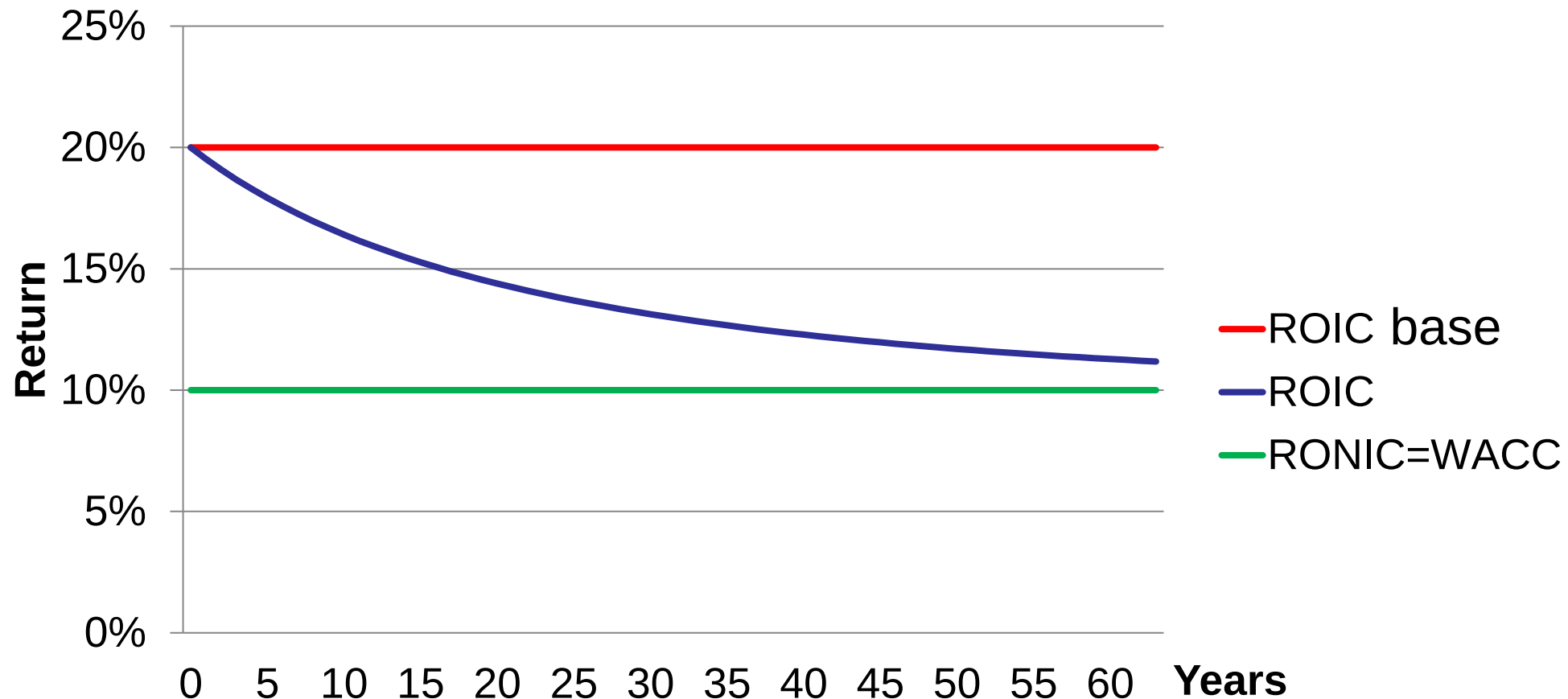
We plan for further details only if those improve the quality of the prediction.

(maintainability and precision of assumptions).

Changes in value creation over time



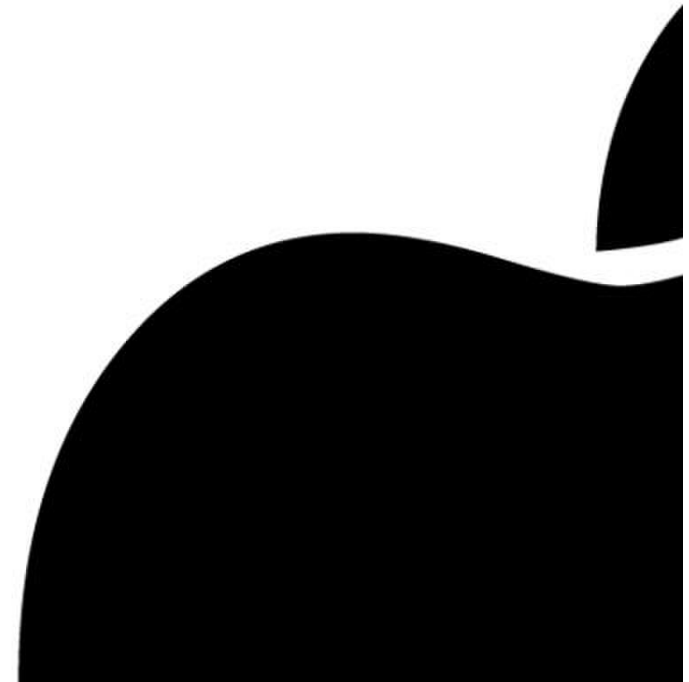
Connections among ROIC, RONIC, and WACC



ROIC remaining high 1

$$\text{ROIC} = (1 - t) * (\text{Unit price} - \text{Unit cost}) * \frac{\text{Quantity}}{\text{IC}}$$

↑
Premium
price?



ROIC remaining high 2

$$\text{ROIC} = (1 - t) * (\text{Unit price} - \text{Unit cost}) * \frac{\text{Quantity}}{\text{IC}}$$

Lower costs?



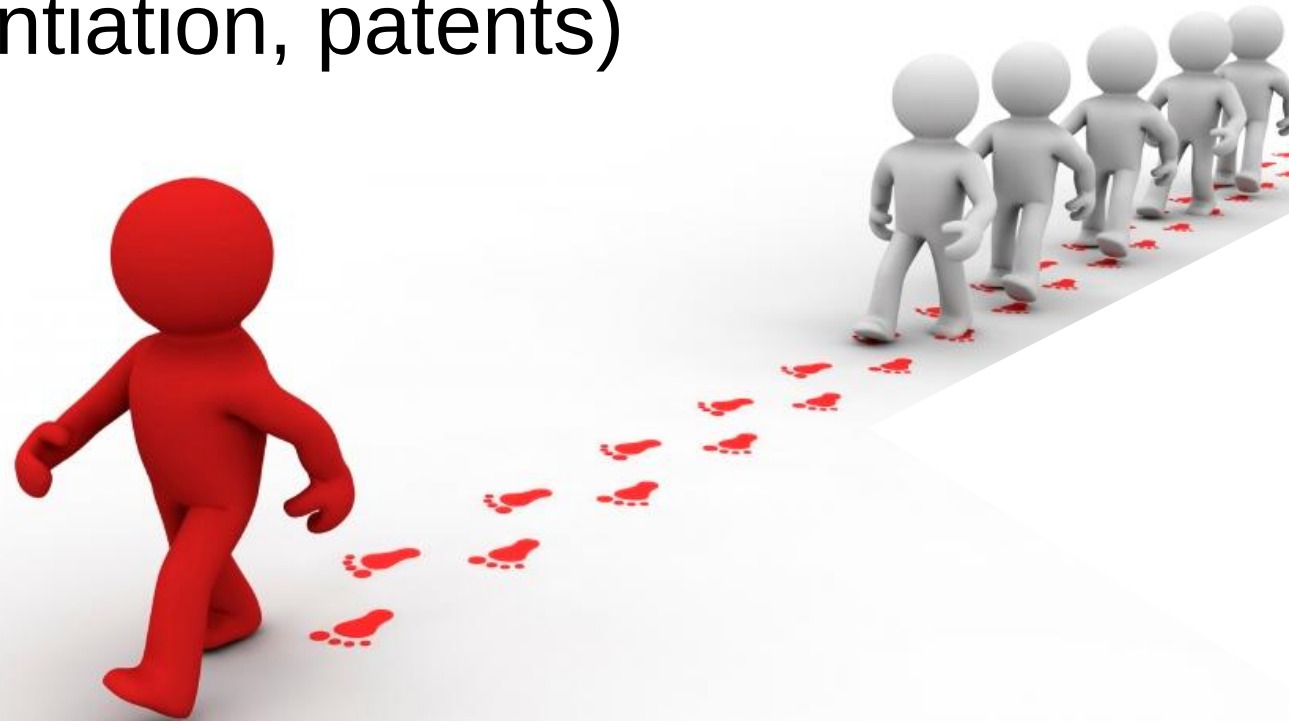
ROIC remaining high 3

$$\text{ROIC} = (1 - t) * (\text{Unit price} - \text{Unit cost}) * \frac{\text{Quantity}}{\text{IC}}$$

More efficient capital
management?



To maintain your competitive advantage in the long run, you have to hinder copying.
(state protection, exclusive contracts, licence, differentiation, patents)



1. FCFF

$$TV = \sum_{t=1}^{\infty} \frac{FCFF_t}{\prod_{i=1}^t (1 + WACC_i)} = \frac{FCFF_{Standard}}{WACC^* - g_{FCFF}^*}$$

Assumption: $WACC^* > g_{FCFF}^*$!

Is that always true?

$$WACC^* < ? > g_{FCFF}^*$$

If $r_f > g_{GDP}$:

$$WACC > r_D > r_f > g_{GDP} \geq g_{Firm}$$

If $WACC < g_{Firm}$, this is not a long-term stable status yet.

$$V_{TV} = \frac{NOPLAT_i - InvCF_i}{WACC^* - g_{NOPLAT}^*} = \frac{NOPLAT(1 - K)}{WACC^* - g_{NOPLAT}^*}$$

$$WACC^* > g_{NOPLAT}^* !$$

$$Inv CF = ?$$

$$= NOPLAT - FCFF = NOPLAT * K =$$

$$= \underline{\text{Net capital investment}} (\Delta IC)$$

$$V_{TV} = \frac{NOPLAT(1 - \frac{g_{NOPLAT}}{RONIC})}{WACC^* - g_{NOPLAT}^*}$$

$WACC^* > g_{NOPLAT}^* < RONIC!$

$WACC^* < ? > RONIC$

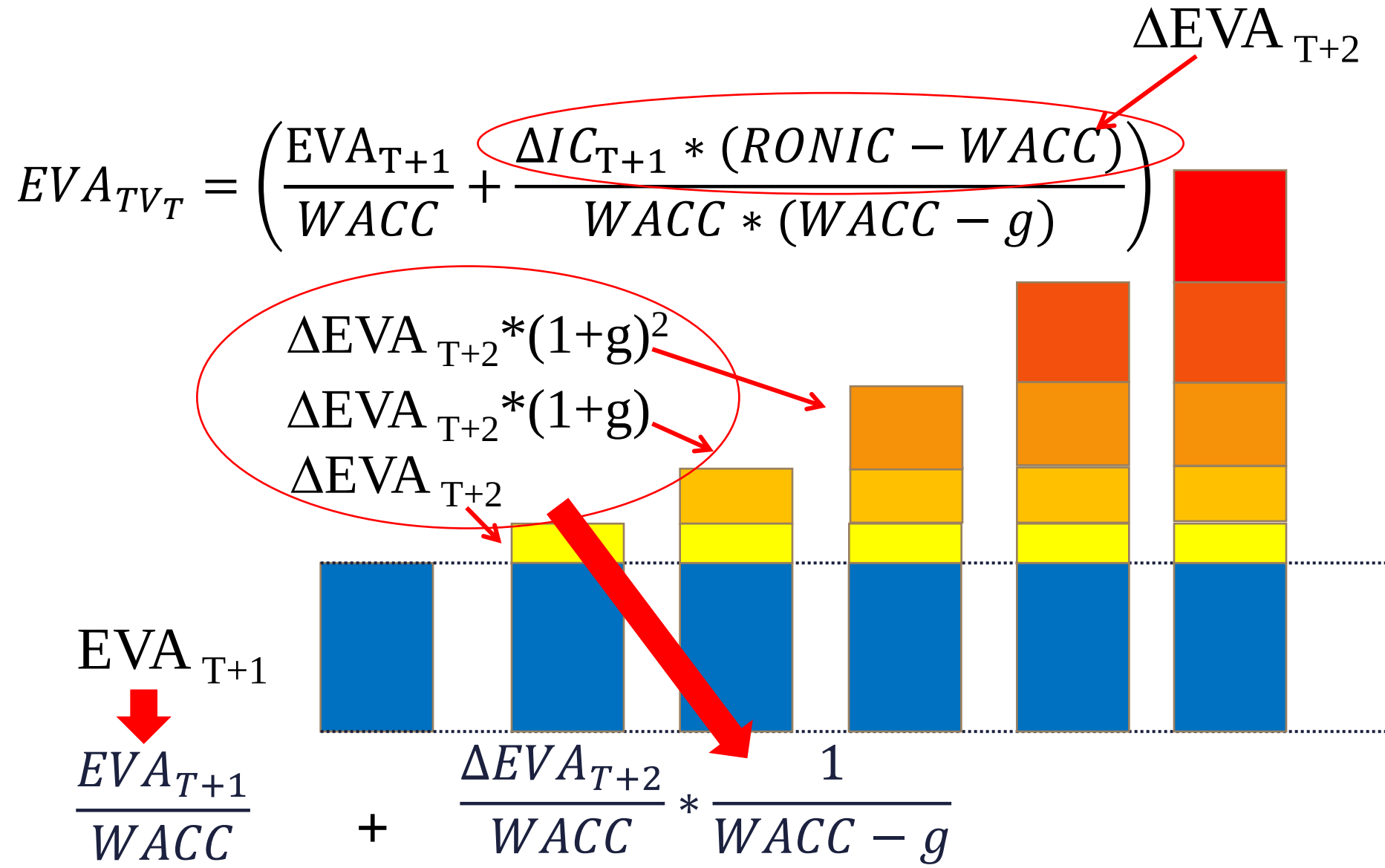
$$TV = \frac{FCFF}{r_A - g} + \frac{D * r_D * t}{r_A - g}$$

$$TV_E = \frac{FCFE}{r_E - g}$$

$$\begin{aligned} EVA &= NOPLAT - IC * WACC \\ &= IC * (ROIC - WACC) \end{aligned}$$

$$EVA_{TV} = \left(\frac{EVA_{T+1}}{WACC} + \frac{NOPLAT_{T+1} * \frac{g}{RONIC} * (RONIC - WACC)}{WACC * (WACC - g)} \right)$$

How did we get that?



Good news: convergence

Due to the competition in the long run: WACC = RONIC

$$V_{TV} = \frac{NOPLAT(1 - \frac{g}{RONIC})}{WACC - g} = \frac{NOPLAT(1 - \frac{g}{WACC})}{WACC - g} =$$

$$\frac{NOPLAT(\frac{WACC}{WACC} - \frac{g}{WACC})}{WACC - g} = \frac{NOPLAT(\frac{WACC - g}{WACC})}{WACC - g} =$$

$$= \frac{NOPLAT}{WACC}$$

Growth (g) plays no role.

What is the meaning of this formula?

$$TV = \frac{NOPLAT}{WACC - g}$$

Growth without investment.

4. Non-CF-based TV methods

Liquidation method

Liquidation or forced liquidation

Replacement methods

What has to and can be replaced from the market?

Multiple-based methods

Data on future market? Future comparable firms?

Typical mistakes in quantifying TV

Wrong choice of starting year

(Cycles, trends, standard operation, first TV year reflect maintainable performance)

Misprediction of ROIC and WACC

(Consider market growth, intensity of competition, and inflation.)

Mistake in discounting

(TV formulas provide the value for the beginning of the TV period.)

Using a real growth rate in a nominal model

(It is harder to believe but has no value effect if $RONIC = WACC$.)

Historic data

Management expectations

Market/industry analysis

Macroanalysis

Predicting growth



What do we predict?

**Sales, operational profitability, EBIT, Noplat,
operational CF, ..., net profit**

Averaging yearly growth rates

Arithmetical or geometric mean?

0	100	
1	120	+20%
2	96	-20%

Arithmetical: $(+20 + -20) / 2 = 0$

Geometrical:

$$\sqrt{(1.2 * 0.8)} = 0.9797 \longrightarrow \text{yearly } -2\%$$

Sustainable growth rate

$$\begin{aligned} g_{\text{Equity}} &= \Delta \text{Equity} / \text{Equity} = \\ &= (\text{Net profit} - \text{Dividend}) / \text{Equity} = \\ &= [\text{Net profit} * (1 - \text{div. payoff ratio})] / \text{Equity} = \\ &= (\text{Net profit} / \text{Equity}) * \text{reinvestment rate} = \\ &= \text{ROE} * \text{reinvestment rate} \end{aligned}$$

Equity and Debt are
book values here!

$$\text{ROE} = \text{ROIC} + \text{Debt/Equity} * [\text{ROIC} - r_D * (1-t)]$$

What changes together with Sales?

Quantity: COGS, variable costs

Receivables, if payment conditions unchanged

Suppliers, as procurement growth (price effect may differ!), if payment terms unchanged

Short term liabilities? Partly. Wages? Taxes?

Debt? Not automatic, a new contract is needed.

Equity: $g = \text{ROE} * \text{reinvestment rate}$

Profit
and CF

Capital
need and
structure

Operational
risk (r_A)

Limits of growth

Can be financed
from inner
sources?

Economic
(GDP/GNP)
growth?

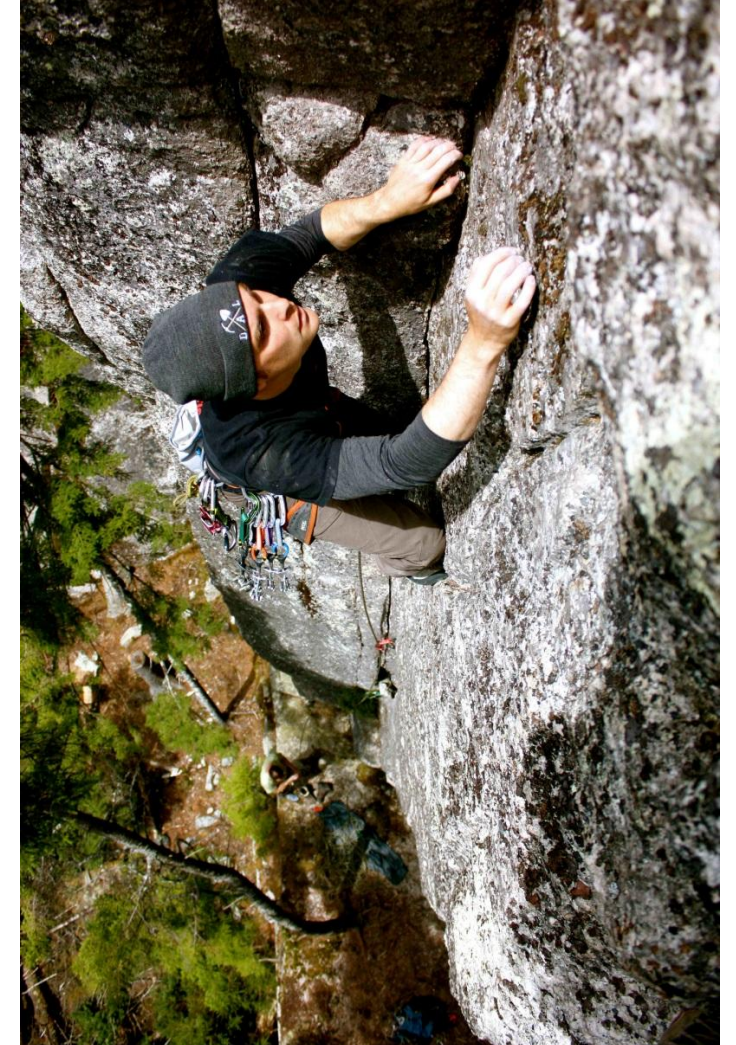
Growth of local
and
international
markets?



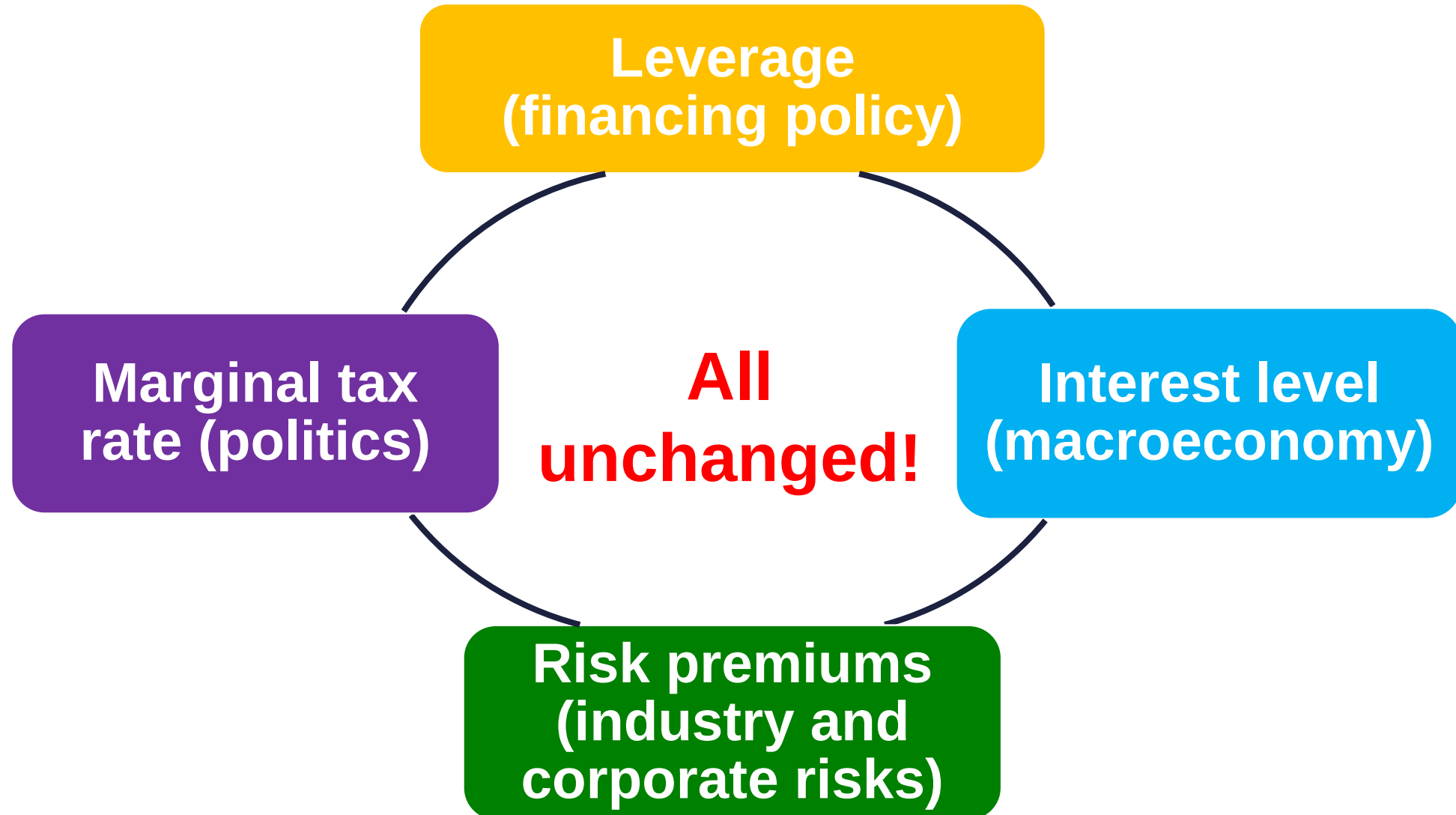
Issues linked to growth

- Nominal or real?
- Inflation stable?
- What currency?
- What supports it?

(market outlook, firm size, current growth, competitive advantages, competitors)



Assumptions of constant WACC



If WACC is constant ...

Leverage is stable and already optimal,

No competitive advantage or drawback,

Industry and regulation unchanged.

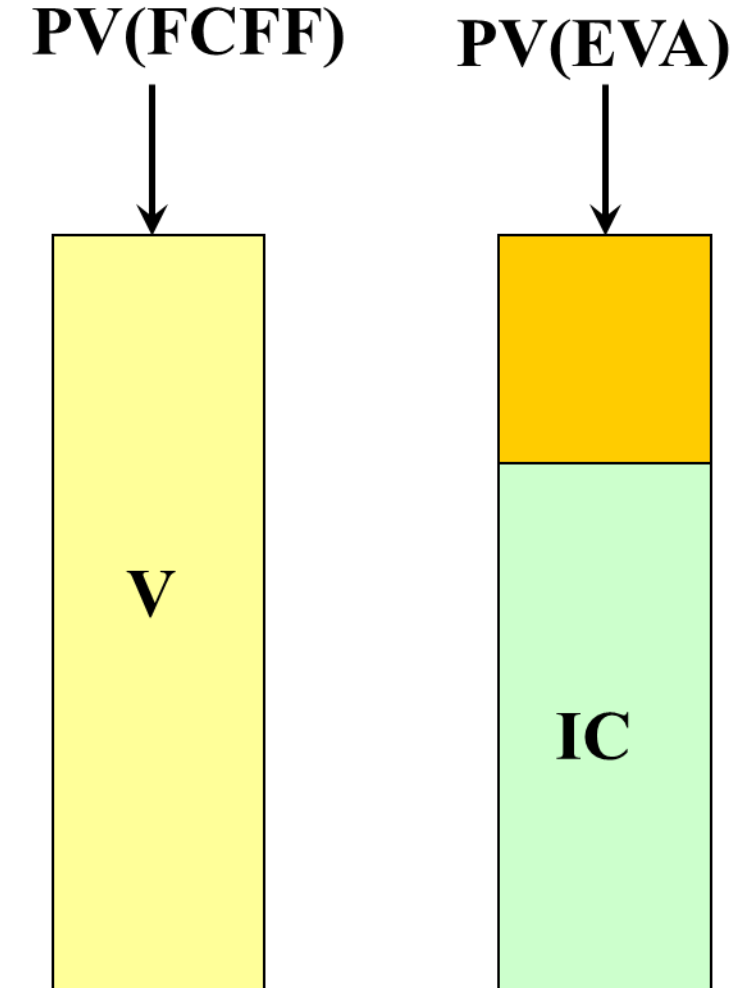


The meaning of TV for different methods

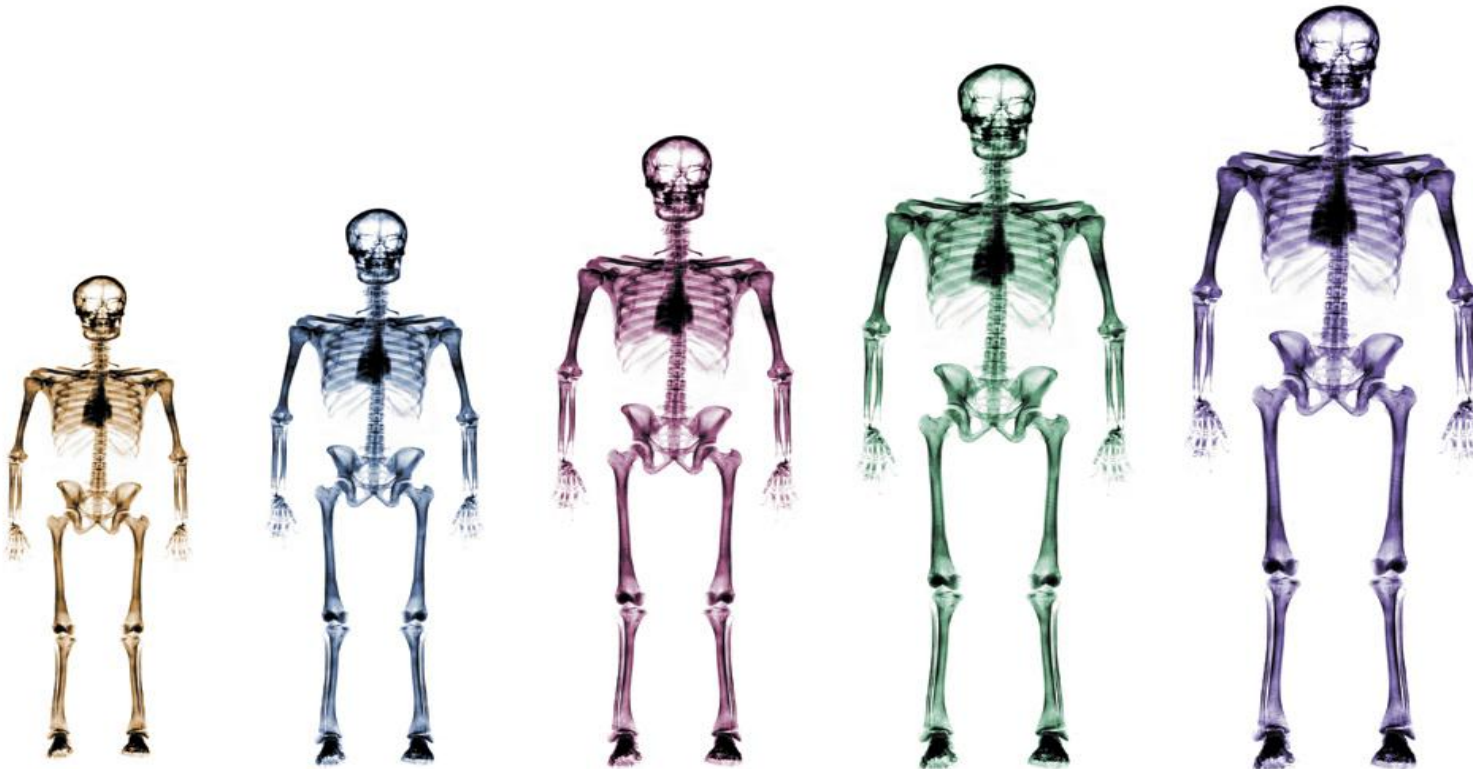
FCFF/FCFE: The value of firm/equity at the start of the TV period.

To get this value, you have to invest IC/book Equity, so that is there without any risk.

EVA/RI: The added value created by the firm during the TV period.



The length of the explicit period should have no value effect. We should extend the explicit period until all inputs reach their stable level.



Even in the case of $RONIC=WACC$ it is not the same what the ROIC is at the beginning of the TV. If $ROIC \gg WACC$, there will be an economic profit generated for very long time.



Lessons of TV 3

Only the TVs of EVA/RI models show what part of the beginning of TV value would be generated during the TV period.



IC



FCFF



EVA/RI

What is important? 1

WACC=10.00%

$IC_{TV}=2000.00$

ROIC	10.00%	10.00%
RONIC	10.00%	10.00%
g	3.00%	7.00%
Noplat	200.00	200.00
K	30.00%	70.00%
FCFF	140.00	60.00
EVA	0.00	0.00
EVA/WACC	0.00	0.00
$TV_{FCFF}=V$	2 000.00	2 000.00
TV_{EVA}	0.00	0.00
TV_{EVA}/V	0.00%	0.00%

What is important? 2

WACC=10.00%

$IC_{TV}=2000.00$

ROIC	15.00%	15.00%
RONIC	10.00%	10.00%
g	3.00%	7.00%
Noplat	300.00	300.00
K	30.00%	70.00%
FCFF	210.00	90.00
EVA	100.00	100.00
EVA/WACC	1 000.00	1 000.00
$TV_{FCFF}=V$	3 000.00	3 000.00
TV_{EVA}	1 000.00	1 000.00
TV_{EVA}/V	33.33%	33.33%

What is important? 3

WACC=10.00%

IC_{TV}=2000.00

ROIC	10.00%	10.00%
RONIC	15.00%	15.00%
g	3.00%	7.00%
Noplat	200.00	200.00
K	20.00%	46.67%
FCFF	160.00	106.67
EVA	0.00	0.00
EVA/WACC	0.00	0.00
TV _{FCFF} =V	2 285.71	3 555.56
TV _{EVA}	285.71	1 555.56
TV _{EVA} /V	12.50%	43.75%

What is important? 4

WACC=10.00%

$IC_{TV}=2000.00$

ROIC	15.00%	15.00%
RONIC	15.00%	15.00%
g	3.00%	7.00%
Noplat	300.00	300.00
K	20.00%	46.67%
FCFF	240.00	160.00
EVA	100.00	100.00
EVA/WACC	1 000.00	1 000.00
$TV_{FCFF}=V$	3 428.57	5 333.33
TV_{EVA}	1 428.57	3 333.33
TV_{EVA}/V	41.67%	62.50%

**Thank you
for your attention.
Any questions?**

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